

SNAPFRAME · DESIGN DECISION RECORD

SnapFrame TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/snapframe		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items D1 to D7); items O1 to O3 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session "/populate to a strong TRL 2") listed nine items as "Proposed, awaiting Amish", and the requirements left the open-gable skin question (R7) without a recommendation. On 2026-09-25 Amish wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided as recommended. Items without a recommendation stay open.

■ Options considered

The options for each item are in the TRL 2 review note and in SNF-PRC-001 v0.2, "Key design choices". They are summarized with each decision below.

■ Decision

Table 1. Decided items.

#	Item	Options	Decision
D1	Budget scope (R10)	(a) \$400 covers the frame kit and tarpaulins come from agency stock; (b) cut cost; (c) raise budget_usd to about \$450	Decided by Amish, 2026-09-25: go with recommendation. (a): the \$400 budget_usd covers the size M frame kit; tarpaulins are agency stock, costed and reported separately. budget_usd stays at 400.
D2	Rafter tube size (R6)	3/4 in throughout (about 18 m/s at TRL 2); 1 in EMT rafters; four frames at 1.33 m	Decided by Amish, 2026-09-25: go with recommendation. 1 in EMT rafters, 3/4 in EMT elsewhere, and the wind rating is stated on the kit.
D3	Node material and process	Printed polymer; cast aluminum	Decided by Amish, 2026-09-25: go with recommendation. Print nodes for the prototype and design every node to be castable from the printed pattern. The polymer itself is not chosen.
D4	Joint locking	Spring buttons only; buttons plus hitch pins	Decided by Amish, 2026-09-25: go with recommendation. Spring buttons at every tube end plus hitch pins at the 12 tension joints (6 feet, 6 eave post sockets).
D5	Frame form	Gable with cable bracing; dome; barrel vault; rigid nodes	Decided by Amish, 2026-09-25: go with recommendation. Gable frame with pinned nodes and cable bracing.
D6	Reference size and family	S 9 m ² , M 16 m ² , L 24 m ²	Decided by Amish, 2026-09-25: go with recommendation. Size family S, M and L, with M as the reference size.
D7	Out of scope	Snow and cyclone rating in the first release	Decided by Amish, 2026-09-25: go with recommendation. Snow and cyclone rating are out of scope for the first release.

Table 2. Items that remain open.

#	Item	Why it is open	Status
O1	Tube offcuts: accept 39 %, shorten posts to 1.50 m, or buy 6 m metric stock	The recommendation was to decide once the first region's tube source is known, so going with it leaves the choice open	Proposed, awaiting Amish
O2	First co-design partner and region	No recommendation was made	Proposed, awaiting Amish
O3	Open gable (R7): two tarpaulins close the roof, both side walls and one gable only; options include a cutting and folding plan, a third tarpaulin, or accepting one open gable as the doorway	No recommendation was made	Proposed, awaiting Amish

■ Consequences

- SNF-REQ-001 v0.3 redefines R10 as the frame kit without tarpaulins. R6 keeps its 20 m/s target.
- SNF-PRC-001 v0.3 and the parametric model use 1 in rafters, which need a second socket bore (30.1 mm) on the eave and ridge nodes.
- SNF-CAL-001 shows that D2 alone does not meet R6: with a first-principles load share, the 3/4 in ridge tubes govern and the rating is about 17.8 m/s. It also shows the frame kit at about \$431 (R10 not met) and the tube bundle at about 27.4 kg (R5 not met). These findings are reported in `docs/REVIEW.md` as new items awaiting Amish; this record does not decide them.
- TRL 4 work is on hold by Amish's instruction.